

PROJECT REPORT

Title: Flask E-Commerce Web Application

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1. INTRODUCTION

The Flask E-Commerce Web Application is a full-stack web development project designed to provide an online shopping platform where users can browse products, add items to a shopping cart, and place orders. The project also includes an admin panel for managing products and monitoring the application. This application is developed using Python Flask framework with database integration and user authentication features.

2. OBJECTIVE

The main objective of this project is to develop a simple and efficient online shopping system that demonstrates real-world e-commerce functionality. The project helps in understanding backend development, frontend integration, authentication systems, and database operations.

3. TECHNOLOGIES USED

Frontend:

- HTML
- CSS

Backend:

- Python
- Flask

Database:

- SQLite

Libraries:

- Flask SQLAlchemy
- Flask Login
- Werkzeug Security

4. FEATURES OF THE PROJECT

User Features:

- User Registration
- User Login and Logout
- Browse Products
- Add Products to Cart
- View Cart

Admin Features:

- Add Products
- Manage Products
- Role-Based Access

System Features:

- Database Integration
- Authentication System
- Responsive User Interface.

5. SYSTEM ARCHITECTURE

The application follows a client-server architecture where:

- Frontend handles user interaction
- Flask backend processes requests
- SQLite database stores user and product data

Flow:

User → Flask Application → Database.

6. MODULES DESCRIPTION

1. Authentication Module

Handles user registration and login functionality.

2. Product Management Module

Allows admin to add and manage products.

3. Shopping Cart Module

Allows users to add products to the cart and calculate totals.

4. Database Module

Stores users, products, and cart information securely.

7. DATABASE TABLES

Users Table:

- ID
- Username
- Password
- Role

Products Table:

- ID
- Product Name
- Price

Cart Table:

- ID

- User ID
- Product Name
- Price

8. WORKING OF THE PROJECT

1. User registers into the system.
2. User logs in using credentials.
3. Products are displayed on the homepage.
4. User adds products to cart.
5. Admin can add and manage products.
6. Cart displays total amount.

9. ADVANTAGES

- Easy to use
- Beginner-friendly project
- Secure login system
- Real-world application structure
- Helps understand full-stack development

10. FUTURE ENHANCEMENTS

- Online Payment Gateway
- Product Images
- Order Tracking
- Email Notifications
- Search and Filter Option

11. CONCLUSION

The Flask E-Commerce Web Application successfully demonstrates the implementation of a basic online shopping platform using Flask and SQLite. The project provides practical knowledge of web development concepts such as authentication, CRUD operations, routing, and database integration. It is a useful project for students to understand real-world web application development.

12. REFERENCES

- Flask Official Documentation
- Python Official Documentation
- SQLite Documentation
- HTML and CSS Tutorials

THANK YOU